

Weekly Progress Log

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 |
Research/Weekly_Progress_Log.md

Summary of work completed each week across all four RUP phases. This log provides evidence of continuous engagement with the project and maps development progress to the planned schedule.

PHASE 1: INCEPTION (Weeks 1–4)

Week 1 — Project Setup and Problem Framing

Dates: 1–8 November 2024

Task	Status	Notes
Project proposal drafted and submitted	Done	Defined problem, proposed solution, initial scope
Initial background reading on Nepal mental health statistics	Done	MoHP 2020 survey; WHO 2021 situational assessment
SuperTokens and eSewa feasibility research	Done	Identified SuperTokens as viable passwordless auth option
Pre-survey questions drafted	Done	10-question Google Form prepared
Technology stack initial selection	Done	Flutter, Node.js, Supabase, SuperTokens, eSewa
GitHub repository created	Done	Private repo; initial Flutter and Node.js project scaffolded

Artefacts: Project proposal; pre-survey draft; technology selection notes

Week 2 — Stakeholder Interviews and Survey Launch

Dates: 9–15 November 2024

Task	Status	Notes
Interview with Ms. Sushana Karki conducted	Done	45-minute semi-structured interview; see 05_Client_Documents
Video interview with Dr. Dibyandra Singh (Zoom)	Done	60-minute call; clinical workflow requirements documented
Pre-survey distributed	Done	WhatsApp groups, university social media, personal network
Risk register drafted	Done	21 risks identified; prioritised by likelihood × impact
Competitor analysis begun	Done	BetterHelp, Headspace, Wysa, HamroPatro review

Artefacts: Interview notes (see 05_Client_Documents); risk register draft; competitor notes

Week 3 — Feasibility Study and Methodology Selection

Dates: 16–22 November 2024

Task	Status	Notes
Pre-survey results collected (166 responses)	Done	Google Forms auto-aggregation; pie chart exports
Feasibility study written	Done	Technical, economic, operational, schedule feasibility
Methodology comparison: Agile vs Waterfall vs RUP vs Prototype	Done	Comparison matrix created; RUP selected
Initial Gantt chart created in Lucidchart	Done	Covers all four RUP phases
Work Breakdown Structure created	Done	Five top-level areas; full task decomposition

Artefacts: Feasibility study; methodology comparison matrix (Appendix E); initial Gantt chart; WBS

Week 4 — Project Environment and Inception Wrap-Up

Dates: 23–30 November 2024

Task	Status	Notes
Flutter project structure established (Clean Architecture)	Done	Feature-first folder structure; placeholder modules created
Node.js project scaffolded	Done	Express app; folder structure; dotenv configured
Supabase project created	Done	Database instance provisioned; initial environment variables set
SuperTokens managed service account created	Done	Passwordless recipe configured in dashboard
Pre-survey findings analysed and documented	Done	Key findings table prepared for report
Inception phase review	Done	All planned artefacts completed

Artefacts: Project scaffolding; environment configuration; survey analysis

PHASE 2: ELABORATION (Weeks 5–8)

Week 5 — Architecture Design and SRS

Dates: 1–7 December 2024

Task	Status	Notes
System architecture diagram created	Done	Three-tier: Flutter → Node.js → Supabase; all external services mapped
SRS drafted	Done	All functional and non-functional requirements documented
Entity-Relationship Diagram drafted	Done	20 tables identified; relationships mapped
Use case diagrams (Customer, Therapist, Admin)	Done	Three separate diagrams; Mermaid source code written

Artefacts: System architecture diagram; SRS (Appendix M); ERD; use case diagrams

Week 6 — SuperTokens and Google Sign-In Prototype

Dates: 8–14 December 2024

Task	Status	Notes
SuperTokens OTP three-step flow prototype built	Done	createCode → consumeCode → /auth/passwordless/complete
Google Sign-In integration prototyped	Done	ThirdParty provider configured in SuperTokens dashboard
Facebook Sign-In integration prototyped	Done	Facebook app credentials configured
JWT middleware for Node.js tested	Done	SuperTokens verifySession() middleware validating tokens
Supabase RLS initial policies written	Done	User self-access policies tested
Wireframes begun in Figma	Done	Onboarding and authentication screens wireframed

Artefacts: SuperTokens prototype; middleware proof-of-concept; initial wireframes

Week 7 — WebRTC Prototype and Figma Mockups

Dates: 15–21 December 2024

Task	Status	Notes
WebRTC peer connection prototype built	Done	flutter_webrtc + Socket.IO signalling; two-device video call achieved
Socket.IO signalling server implemented	Done	join-room, offer, answer, ice-candidate events
STUN server tested with Google STUN	Done	Works on WiFi; fails on some mobile networks — TURN noted as needed
eSewa payment SDK prototype	Done	Test environment payment; success callback handler tested
Figma UI mockups completed	Done	All screens for all three user roles

Task	Status	Notes
Client prototype review — Ms. Karki and Dr. Singh	Done	Feedback incorporated; see 05_Client_Documents
Activity diagrams created	Done	Booking, WebRTC, content publication flows
Sequence diagrams created	Done	OTP login, booking, WebRTC session

Artefacts: WebRTC prototype; eSewa prototype; Figma mockups; activity diagrams; sequence diagrams

Week 8 — Interim Report and Elaboration Wrap-Up

Dates: 22–31 December 2024

Task	Status	Notes
Data Flow Diagrams created	Done	Level 0 and Level 1 for all three user roles
Database schema finalised	Done	All 20 tables; column types; constraints; foreign keys
RLS policies written for all tables	Done	Tested in Supabase dashboard
Interim report written and submitted	Done	Full academic report covering Chapters 1–3
Elaboration phase review	Done	All high-risk components prototyped; architecture validated

Artefacts: DFDs; final database schema; RLS policies; Interim Report submitted

PHASE 3: CONSTRUCTION (Weeks 9–19)

Week 9 — Iteration 1: Authentication Module

Dates: 1–7 January 2025

Task	Status	Notes
SuperTokens OTP flow integrated into Flutter	Done	createCode screen → OTP verification screen → complete
Google Sign-In integrated	Done	flutter_webrtc_google_sign_in package configured
Facebook Sign-In integrated	Done	flutter_facebook_auth configured
Role-based GoRouter navigation guards	Done	'user', 'therapist', 'admin' routes protected
User onboarding preference screens	Done	5-question preference flow; stored in Supabase
GetIt + Injectable DI setup	Done	Service locator registered; all layers wired
Unit tests: TC-U001 to TC-U004 written and passed	Done	OTP registration, duplicate email, Google login, role assignment

Week 10 — Iteration 2: Database Layer and Backend Structure

Dates: 8–14 January 2025

Task	Status	Notes
All 20 database tables created in Supabase	Done	SQL migration scripts written
RLS policies deployed on all tables	Done	Tested with Postman using multiple user tokens
Node.js API structure established	Done	Controllers, services, repositories pattern
Authentication middleware deployed	Done	verifySession() + requireRole() on all protected routes
Base repository classes written	Done	Supabase client queries with error handling
TC-U027, TC-U028 (RLS cross-user tests) written and passed	Done	Confirmed RLS prevents cross-user data access

Week 11 — Iteration 3: Therapist Directory and Admin Verification

Dates: 15–21 January 2025

Task	Status	Notes
Therapist profile creation API	Done	POST /therapists/profile
Admin verification workflow API	Done	PUT /admin/therapists/:id/verify
Therapist directory API with filters	Done	GET /therapists?specialisation=&language;=
Flutter: Therapist directory screen	Done	Filterable card list with fee, specialisation, language
Flutter: Therapist profile page	Done	Full profile with booking button
Flutter: Admin therapist verification screen	Done	Pending applications queue
Unit tests: TC-U023 to TC-U025 written and passed	Done	Profile linking, unverified exclusion, admin verification

Week 12 — Iteration 4: Appointment Booking

Dates: 22–28 January 2025

Task	Status	Notes
Availability slot management API	Done	POST/DELETE /therapists/availability
Appointment creation API	Done	POST /appointments
Real-time slot availability display	Done	Calendar with available/unavailable slots
Double-booking prevention constraint	Done	Unique constraint + backend check
Flutter: Booking calendar screen	Done	Date picker → slot selector → confirm
Unit tests: TC-U005 to TC-U007 written and passed	Done	Valid creation, unverified block, double-booking

Week 13 — Iteration 4 (continued): eSewa Payment Integration

Dates: 29 January – 4 February 2025

Task	Status	Notes
eSewa Flutter SDK integrated	Done	EsewaFlutterSdk.initPayment() wired to booking flow
Payment verification endpoint	Done	POST /payment/verify with HMAC check
Tamper detection for payment callbacks	Done	Amount validation; rejecting modified values
Transaction recording	Done	transactions table populated on confirmed payment
FCM notification on booking confirmation	Done	Patient and therapist both notified
Unit tests: TC-U014 to TC-U016 written and passed	Done	Success callback, failed callback, tampered callback

Week 14 — Iteration 5: WebRTC Video Consultation

Dates: 5–11 February 2025

Task	Status	Notes
Socket.IO signalling server deployed	Done	join-room, offer, answer, ice-candidate events
Flutter WebRTC peer connection	Done	RTCPeerConnection; getUserMedia; remote stream rendering
TURN server deployed	Done	Required for mobile network NAT traversal (R-T01 resolved)
Flutter: Consultation screen (patient view)	Done	Video streams, mute, camera toggle, end session
Flutter: Consultation panel (therapist view)	Done	Video + patient mood chart + PHQ-9/GAD-7 panel
System test: TC-S004, TC-S005 written and passed	Done	Live video call; therapist patient data view

Week 15 — Iteration 5 (continued): Admin Dashboard

Dates: 12–18 February 2025

Task	Status	Notes
Admin analytics API	Done	GET /admin/analytics — user counts, session stats
Admin user management API	Done	GET/DELETE /admin/users
Admin content moderation queue	Done	GET /admin/content/pending; approve/reject actions
Flutter: Admin home dashboard	Done	Statistics cards, recent activity
Flutter: Admin users screen	Done	User list with role filters
Flutter: Content moderation queue screen	Done	Pending articles with approve/reject
System test: TC-S007, TC-S008 written and passed	Done	Content gating; therapist rejection enforcement

Week 16 — Iteration 6: Mood Tracking and Questionnaires

Dates: 19–25 February 2025

Task	Status	Notes
Mood entry API (create + history)	Done	POST /mood; GET /mood?days=7
PHQ-9 submission API	Done	POST /questionnaires/phq9; score calculation; crisis threshold
GAD-7 submission API	Done	POST /questionnaires/gad7
Flutter: Mood logging screen	Done	Slider + emoji labels + note field
Flutter: Mood history chart	Done	fl_chart 7-day and 30-day line charts
Flutter: PHQ-9/GAD-7 questionnaire screens	Done	Per-question answer selection; progress bar
Crisis notice screen	Done	Triggered when PHQ-9 $\geq$ 20; helpline number displayed

Task	Status	Notes
System test: TC-S006 written and passed	Done	PHQ-9 severe threshold triggers crisis notice

Week 17 — Iteration 6 (continued): Content Hub and Cloudinary

Dates: 26 February – 4 March 2025

Task	Status	Notes
Cloudinary upload integrated	Done	Images, audio, video upload from Flutter and Node.js
Content CRUD API	Done	Full create/read/update/delete for all content types
Content hub Flutter screens	Done	Category tabs; article reader; audio player; video player
LRU content cache	Done	Previously loaded content available offline
Subscription-gated premium content	Done	Premium flag check against user's active subscription
System test: TC-S012, TC-S013, TC-S015 written and passed	Done	Subscription gating; audio playback

Week 18 — Iteration 7: Wellness Games and Achievement System

Dates: 5–11 March 2025

Task	Status	Notes
Breathing Bubble game	Done	Animated expand/contract; 4-7-8 breathing pattern
Wellness Trivia game	Done	Multiple choice; score tracking
Affirmation Builder game	Done	Selectable affirmations; personalisation
Motivational Story game	Done	Narrative choice mechanics
Game session recording API	Done	POST /games/sessions

Task	Status	Notes
Achievement unlock logic	Done	Event-triggered checks; badge awarded on first occurrence
Flutter: Achievement gallery	Done	Profile page with unlocked badges
Unit tests: TC-U019, TC-U020 written and passed	Done	Session recording; achievement award

Week 19 — Iteration 7 (continued): Offline Sync, Push Notifications, Subscriptions

Dates: 12–18 March 2025

Task	Status	Notes
SQLite offline mood logging	Done	sqlite integration; background sync on connectivity restore
Offline-to-online sync job	Done	connectivity_plus listener → flush queue on reconnect
FCM push notifications (all event types)	Done	Booking confirmation, session reminder, content approval
Subscription creation API	Done	POST /subscriptions; plan tier gating
Flutter: Subscription plans screen	Done	Free/Silver/Gold tier display
Unit tests: TC-U029, TC-U030 written and passed	Done	Offline sync; cache serving
Final Construction phase review	Done	All 7 iterations complete; full feature set delivered

PHASE 4: TRANSITION (Week 20)

Week 20 — Testing, UAT, Post-Survey, Final Report

Dates: 19 March – 30 April 2025

Task	Status	Notes
System test cases TC-S001 to TC-S015 executed	Done	All 15 passed
UAT planning and participant recruitment	Done	5 target users + 2 clients identified and scheduled
UAT sessions conducted (5 target users)	Done	TC-UAT001 to TC-UAT013, TC-UAT021 to TC-UAT025
UAT sessions conducted (Ms. Karki — therapist role)	Done	TC-UAT014 to TC-UAT016, TC-UAT020
UAT sessions conducted (Dr. Singh — admin role)	Done	TC-UAT017 to TC-UAT019
Post-survey distributed and collected	Done	132 responses; 100% positive outcomes
User feedback forms collected	Done	All 5 target users + 2 clients
Client approval letters obtained	Done	Signed by Ms. Karki and Dr. Singh
System screenshots captured	Done	All major screens; see Appendix J.12
Final Gantt chart updated with actual timeline	Done	Deviations noted: WebRTC +1 week; eSewa -1 week
FYP report finalised	Done	All 6 chapters + appendices A–N
Report submitted	Done	Submitted April 2025

CONSTRUCTION ITERATION SUMMARY

Iteration	Weeks	Primary Deliverable	Status
1	9–10	Authentication + onboarding	Complete
2	10–11	Database layer + API structure	Complete
3	11–12	Therapist directory + admin verification	Complete

Iteration	Weeks	Primary Deliverable	Status
4	12–13	Appointment booking + eSewa payment	Complete
5	14–15	WebRTC consultation + admin dashboard	Complete
6	16–17	Mood tracking + questionnaires + content hub	Complete
7	18–19	Wellness games + achievements + offline + notifications	Complete